

Welcome to the minitutorial!

**To encourage interaction, please come to the ground floor
and site near the front row.**

<https://github.com/PerformanceEstimation/Tutorial-SIAM-OP26>



Performance Estimation Problems (PEPs)

Systematic analysis and design of first-order optimization algorithms
via the performance estimation framework

François Glineur, Adrien Taylor



SIAM Conference on Optimization, tutorial session – June 2026



PEPs on an example: gradient descent

PEPs and proof structures

PEPs to design a (surprising?) algorithm

PEPs on an example: gradient descent

PEPs and proof structures

PEPs to design a (surprising?) algorithm

| Concrete example: analysis of a gradient step

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ (continuously differentiable). Find $x_\star \in \mathbb{R}^d$ such that

$$f(x_\star) = \min_{x \in \mathbb{R}^d} f(x),$$

with f is L -smooth and μ -strongly convex ($f \in \mathcal{F}_{\mu,L}$).

| Concrete example: analysis of a gradient step

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ (continuously differentiable). Find $x_\star \in \mathbb{R}^d$ such that

$$f(x_\star) = \min_{x \in \mathbb{R}^d} f(x),$$

with f is L -smooth and μ -strongly convex ($f \in \mathcal{F}_{\mu,L}$).

(Gradient method) We decide to use: $x_{k+1} = x_k - \gamma \nabla f(x_k)$

| Concrete example: analysis of a gradient step

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ (continuously differentiable). Find $x_\star \in \mathbb{R}^d$ such that

$$f(x_\star) = \min_{x \in \mathbb{R}^d} f(x),$$

with f is L -smooth and μ -strongly convex ($f \in \mathcal{F}_{\mu,L}$).

(Gradient method) We decide to use: $x_{k+1} = x_k - \gamma \nabla f(x_k)$

Question: what *a priori* guarantees after N iterations?

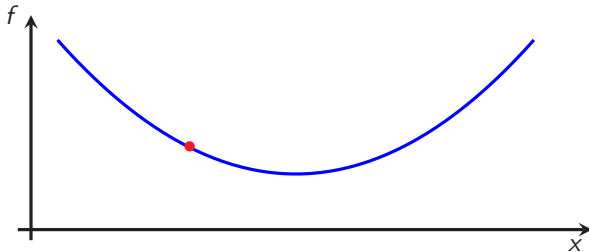
Examples: what about $f(x_N) - f(x_\star)$, $\|\nabla f(x_N)\|$, $\|x_N - x_\star\|$?

| About the assumptions

A differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is μ -strongly convex and L -smooth iff $\forall x, y \in \mathbb{R}^d$:

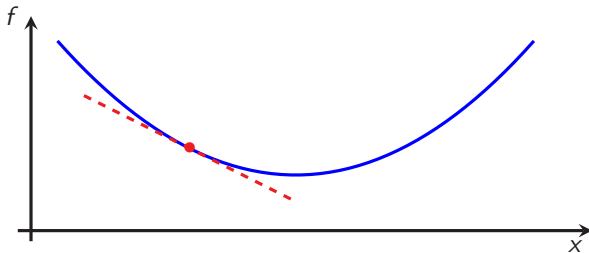
About the assumptions

A differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is μ -strongly convex and L -smooth iff $\forall x, y \in \mathbb{R}^d$:



About the assumptions

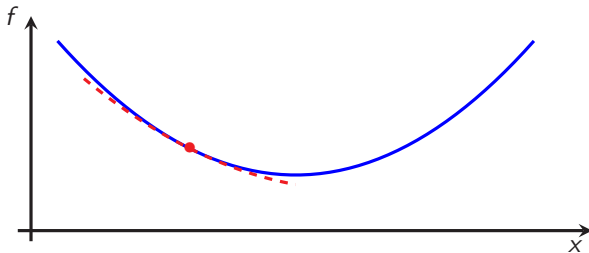
A differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is μ -strongly convex and L -smooth iff $\forall x, y \in \mathbb{R}^d$:



(1) (Convexity) $f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle$,

About the assumptions

A differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is μ -strongly convex and L -smooth iff $\forall x, y \in \mathbb{R}^d$:

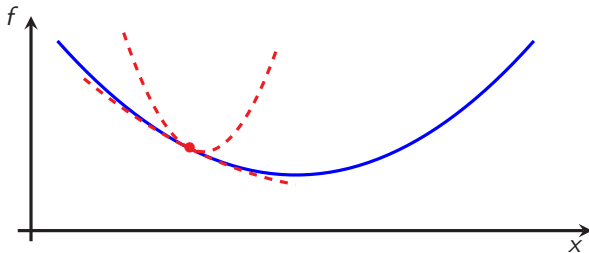


(1) (Convexity) $f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle$,

(1b) (μ -strong convexity) $f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle + \frac{\mu}{2} \|x - y\|^2$,

About the assumptions

A differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is μ -strongly convex and L -smooth iff $\forall x, y \in \mathbb{R}^d$:



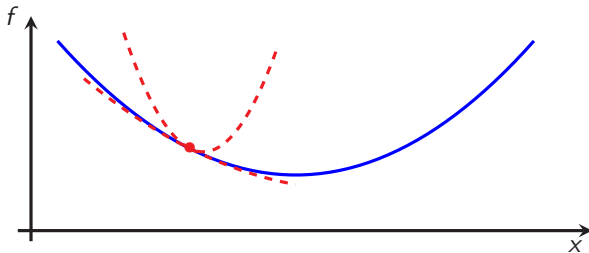
(1) (Convexity) $f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle$,

(1b) (μ -strong convexity) $f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle + \frac{\mu}{2} \|x - y\|^2$,

(2) (L -smoothness) $f(x) \leq f(y) + \langle \nabla f(y), x - y \rangle + \frac{L}{2} \|x - y\|^2$,

About the assumptions

A differentiable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is μ -strongly convex and L -smooth iff $\forall x, y \in \mathbb{R}^d$:



(1) (Convexity) $f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle$,

(1b) (μ -strong convexity) $f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle + \frac{\mu}{2} \|x - y\|^2$,

(2) (L -smoothness) $f(x) \leq f(y) + \langle \nabla f(y), x - y \rangle + \frac{L}{2} \|x - y\|^2$,

(1&2) $\langle \nabla f(x) - \nabla f(y); x - y \rangle \geq \frac{1}{L+\mu} \|\nabla f(x) - \nabla f(y)\|^2 + \frac{\mu L}{L+\mu} \|x - y\|^2$.

| Convergence rate of a gradient step

| Convergence rate of a gradient step

Toy example: What is the smallest τ such that:

$$\|x_1 - x_\star\|^2 \leq \tau \|x_0 - x_\star\|^2,$$

for all

- ◇ $d \in \mathbb{N}$, L -smooth and μ -strongly convex function f (notation $f \in \mathcal{F}_{\mu,L}$),
- ◇ x_0 , and x_1 generated by gradient step $x_1 = x_0 - \gamma \nabla f(x_0)$,
- ◇ $x_\star = \underset{x}{\operatorname{argmin}} f(x)$?

| Convergence rate of a gradient step

Toy example: What is the smallest τ such that:

$$\|x_1 - x_\star\|^2 \leq \tau \|x_0 - x_\star\|^2,$$

for all

- ◇ $d \in \mathbb{N}$, L -smooth and μ -strongly convex function f (notation $f \in \mathcal{F}_{\mu,L}$),
- ◇ x_0 , and x_1 generated by gradient step $x_1 = x_0 - \gamma \nabla f(x_0)$,
- ◇ $x_\star = \underset{x}{\operatorname{argmin}} f(x)$?

Computing τ ?¹

¹Original idea from [Drori and Teboulle, 2014]. Developments here from [T, Hendrickx, Glineur, 2017].

| Convergence rate of a gradient step

Toy example: What is the smallest τ such that:

$$\|x_1 - x_\star\|^2 \leq \tau \|x_0 - x_\star\|^2,$$

for all

- ◇ $d \in \mathbb{N}$, L -smooth and μ -strongly convex function f (notation $f \in \mathcal{F}_{\mu,L}$),
- ◇ x_0 , and x_1 generated by gradient step $x_1 = x_0 - \gamma \nabla f(x_0)$,
- ◇ $x_\star = \underset{x}{\operatorname{argmin}} f(x)$?

Computing τ ?¹

$$\tau = \max_{f, x_0, x_1, x_\star} \frac{\|x_1 - x_\star\|^2}{\|x_0 - x_\star\|^2}$$

s.t. $f \in \mathcal{F}_{\mu,L}$

Functional class

¹Original idea from [Drori and Teboulle, 2014]. Developments here from [T, Hendrickx, Glineur, 2017].

| Convergence rate of a gradient step

Toy example: What is the smallest τ such that:

$$\|x_1 - x_\star\|^2 \leq \tau \|x_0 - x_\star\|^2,$$

for all

- ◇ $d \in \mathbb{N}$, L -smooth and μ -strongly convex function f (notation $f \in \mathcal{F}_{\mu,L}$),
- ◇ x_0 , and x_1 generated by gradient step $x_1 = x_0 - \gamma \nabla f(x_0)$,
- ◇ $x_\star = \underset{x}{\operatorname{argmin}} f(x)$?

Computing τ ?¹

$$\tau = \max_{f, x_0, x_1, x_\star} \frac{\|x_1 - x_\star\|^2}{\|x_0 - x_\star\|^2}$$

$$\text{s.t. } f \in \mathcal{F}_{\mu,L}$$

$$x_1 = x_0 - \gamma \nabla f(x_0)$$

$$\nabla f(x_\star) = 0$$

Functional class

Algorithm

Optimality of $x_\star$

¹Original idea from [Drori and Teboulle, 2014]. Developments here from [T, Hendrickx, Glineur, 2017].

| Convergence rate of a gradient step

Toy example: What is the smallest τ such that:

$$\|x_1 - x_\star\|^2 \leq \tau \|x_0 - x_\star\|^2,$$

for all

- ◇ $d \in \mathbb{N}$, L -smooth and μ -strongly convex function f (notation $f \in \mathcal{F}_{\mu,L}$),
- ◇ x_0 , and x_1 generated by gradient step $x_1 = x_0 - \gamma \nabla f(x_0)$,
- ◇ $x_\star = \underset{x}{\operatorname{argmin}} f(x)$?

Computing τ ?¹

$$\tau = \max_{f, x_0, x_1, x_\star} \frac{\|x_1 - x_\star\|^2}{\|x_0 - x_\star\|^2}$$

$$\text{s.t. } f \in \mathcal{F}_{\mu,L}$$

Functional class

Variables: $f, x_0, x_1, x_\star$.

$$x_1 = x_0 - \gamma \nabla f(x_0)$$

Algorithm

Parameters: μ, L, γ .

$$\nabla f(x_\star) = 0$$

Optimality of $x_\star$

¹Original idea from [Drori and Teboulle, 2014]. Developments here from [T, Hendrickx, Glineur, 2017].

| Replace constraints

| Replace constraints

- ◇ Interpolation conditions allow removing **red** constraints

$$\begin{aligned} & \max_{\substack{x_0, x_1, x_\star \\ g_0, g_\star \\ f_0, f_\star}} \frac{\|x_1 - x_\star\|^2}{\|x_0 - x_\star\|^2} \\ & \text{subject to} \quad \exists f \in \mathcal{F}_{\mu, L} \text{ such that } \begin{cases} f_i = f(x_i) & i = 0, \star \\ g_i = \nabla f(x_i) & i = 0, \star \end{cases} \\ & \quad x_1 = x_0 - \gamma g_0 \\ & \quad g_\star = 0, \end{aligned}$$

| Replace constraints

- ◇ Interpolation conditions allow removing **red** constraints

$$\begin{aligned} & \max_{\substack{x_0, x_1, x_\star \\ g_0, g_\star \\ f_0, f_\star}} \frac{\|x_1 - x_\star\|^2}{\|x_0 - x_\star\|^2} \\ & \text{subject to} \quad \exists f \in \mathcal{F}_{\mu, L} \text{ such that } \begin{cases} f_i = f(x_i) & i = 0, \star \\ g_i = \nabla f(x_i) & i = 0, \star \end{cases} \\ & \quad x_1 = x_0 - \gamma g_0 \\ & \quad g_\star = 0, \end{aligned}$$

- ◇ replacing them by

$$\begin{aligned} f_\star &\geq f_0 + \langle g_0, x_\star - x_0 \rangle + \frac{1}{2L} \|g_\star - g_0\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_\star - x_0 - \frac{1}{L}(g_\star - g_0)\|^2 \\ f_0 &\geq f_\star + \langle g_\star, x_0 - x_\star \rangle + \frac{1}{2L} \|g_0 - g_\star\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_0 - x_\star - \frac{1}{L}(g_0 - g_\star)\|^2. \end{aligned}$$

| Replace constraints

- ◇ Interpolation conditions allow removing **red** constraints

$$\begin{aligned} & \max_{\substack{x_0, x_1, x_\star \\ g_0, g_\star \\ f_0, f_\star}} \frac{\|x_1 - x_\star\|^2}{\|x_0 - x_\star\|^2} \\ & \text{subject to} \quad \exists f \in \mathcal{F}_{\mu, L} \text{ such that } \begin{cases} f_i = f(x_i) & i = 0, \star \\ g_i = \nabla f(x_i) & i = 0, \star \end{cases} \\ & \quad x_1 = x_0 - \gamma g_0 \\ & \quad g_\star = 0, \end{aligned}$$

- ◇ replacing them by

$$\begin{aligned} f_\star &\geq f_0 + \langle g_0, x_\star - x_0 \rangle + \frac{1}{2L} \|g_\star - g_0\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_\star - x_0 - \frac{1}{L}(g_\star - g_0)\|^2 \\ f_0 &\geq f_\star + \langle g_\star, x_0 - x_\star \rangle + \frac{1}{2L} \|g_0 - g_\star\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_0 - x_\star - \frac{1}{L}(g_0 - g_\star)\|^2. \end{aligned}$$

- ◇ Same optimal value (no relaxation): **non-convex quadratic** problem.

| Semidefinite lifting

◇ Define $P \triangleq [x_0 - x_*, g_0] \in \mathbb{R}^{d \times 2}$ and $F \triangleq f_0 - f_*$.

| Semidefinite lifting

- ◇ Define $P \triangleq [x_0 - x_*, g_0] \in \mathbb{R}^{d \times 2}$ and $F \triangleq f_0 - f_*$.
- ◇ Using new variables $G \succcurlyeq 0$ and F

$$G \triangleq P^T P = \begin{bmatrix} \|x_0 - x_*\|^2 & \langle g_0, x_0 - x_* \rangle \\ \langle g_0, x_0 - x_* \rangle & \|g_0\|^2 \end{bmatrix} \succcurlyeq 0,$$

| Semidefinite lifting

◇ Define $P \triangleq [x_0 - x_*, g_0] \in \mathbb{R}^{d \times 2}$ and $F \triangleq f_0 - f_*$.

◇ Using new variables $G \succcurlyeq 0$ and F

$$G \triangleq P^T P = \begin{bmatrix} \|x_0 - x_*\|^2 & \langle g_0, x_0 - x_* \rangle \\ \langle g_0, x_0 - x_* \rangle & \|g_0\|^2 \end{bmatrix} \succcurlyeq 0,$$

◇ previous problem can be relaxed to 2×2 SDP

$$\begin{aligned} \max_{G, F} \quad & G_{1,1} + \gamma^2 G_{2,2} - 2\gamma G_{1,2} \\ \text{subject to} \quad & F + \frac{L\mu}{2(L-\mu)} G_{1,1} + \frac{1}{2(L-\mu)} G_{2,2} - \frac{L}{L-\mu} G_{1,2} \leq 0 \\ & -F + \frac{L\mu}{2(L-\mu)} G_{1,1} + \frac{1}{2(L-\mu)} G_{2,2} - \frac{\mu}{L-\mu} G_{1,2} \leq 0 \\ & G_{1,1} = 1 \\ & G \succcurlyeq 0 \end{aligned}$$

| Semidefinite lifting

◇ Define $P \triangleq [x_0 - x_*, g_0] \in \mathbb{R}^{d \times 2}$ and $F \triangleq f_0 - f_*$.

◇ Using new variables $G \succcurlyeq 0$ and F

$$G \triangleq P^T P = \begin{bmatrix} \|x_0 - x_*\|^2 & \langle g_0, x_0 - x_* \rangle \\ \langle g_0, x_0 - x_* \rangle & \|g_0\|^2 \end{bmatrix} \succcurlyeq 0,$$

◇ previous problem can be relaxed to 2×2 SDP

$$\begin{aligned} \max_{G, F} \quad & G_{1,1} + \gamma^2 G_{2,2} - 2\gamma G_{1,2} \\ \text{subject to} \quad & F + \frac{L\mu}{2(L-\mu)} G_{1,1} + \frac{1}{2(L-\mu)} G_{2,2} - \frac{L}{L-\mu} G_{1,2} \leq 0 \\ & -F + \frac{L\mu}{2(L-\mu)} G_{1,1} + \frac{1}{2(L-\mu)} G_{2,2} - \frac{\mu}{L-\mu} G_{1,2} \leq 0 \\ & G_{1,1} = 1 \\ & G \succcurlyeq 0 \end{aligned}$$

(using homogeneity argument and substituting x_1 and g_*).

| Semidefinite lifting

◇ Define $P \triangleq [x_0 - x_*, g_0] \in \mathbb{R}^{d \times 2}$ and $F \triangleq f_0 - f_*$.

◇ Using new variables $G \succcurlyeq 0$ and F

$$G \triangleq P^T P = \begin{bmatrix} \|x_0 - x_*\|^2 & \langle g_0, x_0 - x_* \rangle \\ \langle g_0, x_0 - x_* \rangle & \|g_0\|^2 \end{bmatrix} \succcurlyeq 0,$$

◇ previous problem can be relaxed to 2×2 SDP

$$\begin{aligned} \max_{G, F} \quad & G_{1,1} + \gamma^2 G_{2,2} - 2\gamma G_{1,2} \\ \text{subject to} \quad & F + \frac{L\mu}{2(L-\mu)} G_{1,1} + \frac{1}{2(L-\mu)} G_{2,2} - \frac{L}{L-\mu} G_{1,2} \leq 0 \\ & -F + \frac{L\mu}{2(L-\mu)} G_{1,1} + \frac{1}{2(L-\mu)} G_{2,2} - \frac{\mu}{L-\mu} G_{1,2} \leq 0 \\ & G_{1,1} = 1 \\ & G \succcurlyeq 0 \end{aligned}$$

(using homogeneity argument and substituting x_1 and g_*).

◇ Assuming $x_0, x_*, g_0 \in \mathbb{R}^d$ with $d \geq 2$, same optimal value as original problem!

| Interactive parts begin!



<https://github.com/PerformanceEstimation/Tutorial-SIAM-OP26>

| Numerical worst-case bound from the SDP

Set $L = 1$, $\mu = .1$ and solve SDP for a few γ 's.



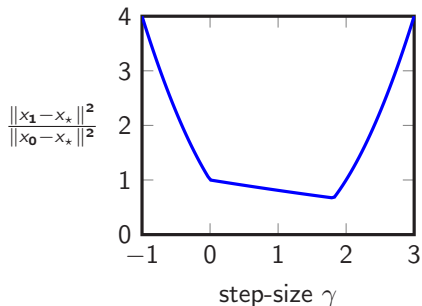
Your turn! Complete and run the corresponding notebook cells.

Numerical worst-case bound from the SDP

Set $L = 1$, $\mu = .1$ and solve SDP for a few γ 's.



Your turn! Complete and run the corresponding notebook cells.

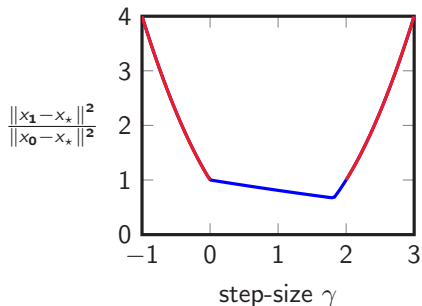


Numerical worst-case bound from the SDP

Set $L = 1$, $\mu = .1$ and solve SDP for a few γ 's.



Your turn! Complete and run the corresponding notebook cells.



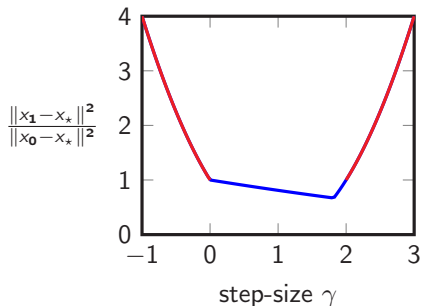
Can you guess the closed form?

Numerical worst-case bound from the SDP

Set $L = 1$, $\mu = .1$ and solve SDP for a few γ 's.



Your turn! Complete and run the corresponding notebook cells.



Can you guess the closed form? Observation: numerics match the known $\max\{(1 - \gamma L)^2, (1 - \gamma \mu)^2\}$.

| Dual problem

◇ Primal problem (abstract form)

$$\begin{aligned} \max_{G, F} \quad & \langle b_{\text{obj}}, F \rangle + \text{Tr}(A_{\text{obj}} G) \quad \text{subject to:} \quad \langle b_1, F \rangle + \text{Tr}(A_1 G) \leq 0 \\ & \langle b_2, F \rangle + \text{Tr}(A_2 G) \leq 0 \\ & \langle b_{\text{init}}, F \rangle + \text{Tr}(A_{\text{init}} G) = 1 \\ & G \succcurlyeq 0 \end{aligned}$$

| Dual problem

◇ Primal problem (abstract form)

$$\begin{array}{ll} \max_{G, F} \langle b_{\text{obj}}, F \rangle + \text{Tr}(A_{\text{obj}} G) & \text{subject to:} \\ \langle b_1, F \rangle + \text{Tr}(A_1 G) \leq 0 & : \lambda_1 \geq 0 \\ \langle b_2, F \rangle + \text{Tr}(A_2 G) \leq 0 & : \lambda_2 \geq 0 \\ \langle b_{\text{init}}, F \rangle + \text{Tr}(A_{\text{init}} G) = 1 & : \tau \\ G \succcurlyeq 0 & : S \succcurlyeq 0 \end{array}$$

| Dual problem

- ◇ Primal problem (abstract form)

$$\begin{aligned} \max_{G, F} \langle b_{\text{obj}}, F \rangle + \text{Tr}(A_{\text{obj}} G) \quad \text{subject to:} \quad & \langle b_1, F \rangle + \text{Tr}(A_1 G) \leq 0 & : \lambda_1 \geq 0 \\ & \langle b_2, F \rangle + \text{Tr}(A_2 G) \leq 0 & : \lambda_2 \geq 0 \\ & \langle b_{\text{init}}, F \rangle + \text{Tr}(A_{\text{init}} G) = 1 & : \tau \\ & G \succcurlyeq 0 & : S \succcurlyeq 0 \end{aligned}$$

- ◇ Lagrangian relaxation (provides upper bounds for any $\lambda_1, \lambda_2 \geq 0, S \succcurlyeq 0, \tau \in \mathbb{R}$):

$$\max_{G, F} \tau + \langle [b_{\text{obj}} - \lambda_1 b_1 - \lambda_2 b_2 - \tau b_{\text{init}}], F \rangle + \text{Tr}([A_{\text{obj}} - \lambda_1 A_1 - \lambda_2 A_2 - \tau A_{\text{init}} + S] G)$$

| Dual problem

- ◇ Primal problem (abstract form)

$$\begin{aligned} \max_{G, F} \langle b_{\text{obj}}, F \rangle + \text{Tr}(A_{\text{obj}} G) \quad \text{subject to:} \quad & \langle b_1, F \rangle + \text{Tr}(A_1 G) \leq 0 & : \lambda_1 \geq 0 \\ & \langle b_2, F \rangle + \text{Tr}(A_2 G) \leq 0 & : \lambda_2 \geq 0 \\ & \langle b_{\text{init}}, F \rangle + \text{Tr}(A_{\text{init}} G) = 1 & : \tau \\ & G \succcurlyeq 0 & : S \succcurlyeq 0 \end{aligned}$$

- ◇ Lagrangian relaxation (provides upper bounds for any $\lambda_1, \lambda_2 \geq 0, S \succcurlyeq 0, \tau \in \mathbb{R}$):

$$\max_{G, F} \tau + \langle [b_{\text{obj}} - \lambda_1 b_1 - \lambda_2 b_2 - \tau b_{\text{init}}], F \rangle + \text{Tr}([A_{\text{obj}} - \lambda_1 A_1 - \lambda_2 A_2 - \tau A_{\text{init}} + S] G)$$

- ◇ Dual problem (minimize upper bound)

$$\begin{aligned} \min_{\substack{\lambda_1, \lambda_2 \geq 0 \\ S \succcurlyeq 0 \\ \tau \in \mathbb{R}}} \tau \quad \text{subject to:} \quad & 0 = b_{\text{obj}} - \lambda_1 b_1 - \lambda_2 b_2 - \tau b_{\text{init}} \\ & S = \lambda_1 A_1 + \lambda_2 A_2 + \tau A_{\text{init}} - A_{\text{obj}} \end{aligned}$$

| Dual solutions (numerical) — back to gradient descent

Set $L = 1$, $\mu = .1$ and recover λ_1, λ_2 for a few γ 's.



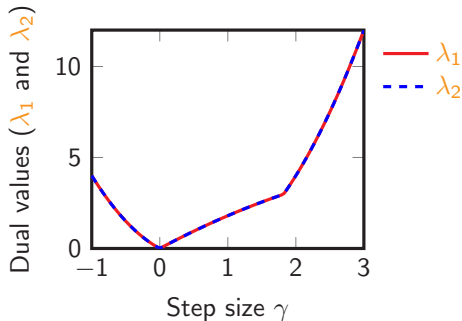
Your turn! Complete and run the corresponding notebook cells.

| Dual solutions (numerical) — back to gradient descent

Set $L = 1$, $\mu = .1$ and recover λ_1, λ_2 for a few γ 's.



Your turn! Complete and run the corresponding notebook cells.

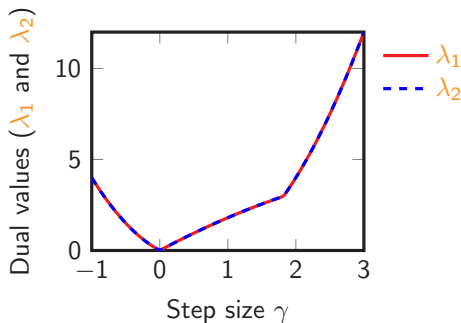


| Dual solutions (numerical) — back to gradient descent

Set $L = 1$, $\mu = .1$ and recover λ_1, λ_2 for a few γ 's.



Your turn! Complete and run the corresponding notebook cells.



Numerics match $\lambda_1 = \lambda_2 = 2|\gamma|\rho(\gamma)$ with $\rho(\gamma) = \max\{\gamma L - 1, 1 - \gamma\mu\}$.

| Dual problem — back to gradient descent

| Dual problem — back to gradient descent

◇ Dual problem is

$$\min_{\tau, \lambda_1, \lambda_2 \geq 0} \tau \quad \text{subject to} \quad S = \begin{bmatrix} \tau - 1 + \frac{\lambda_1 L \mu}{L - \mu} & \gamma - \frac{\lambda_1 (L + \mu)}{2(L - \mu)} \\ \gamma - \frac{\lambda_1 (L + \mu)}{2(L - \mu)} & \frac{\lambda_1}{L - \mu} - \gamma^2 \end{bmatrix} \succcurlyeq 0$$
$$0 = \lambda_1 - \lambda_2.$$

| Dual problem — back to gradient descent

◇ Dual problem is

$$\min_{\tau, \lambda_1, \lambda_2 \geq 0} \tau \quad \text{subject to} \quad S = \begin{bmatrix} \tau - 1 + \frac{\lambda_1 L \mu}{L - \mu} & \gamma - \frac{\lambda_1 (L + \mu)}{2(L - \mu)} \\ \gamma - \frac{\lambda_1 (L + \mu)}{2(L - \mu)} & \frac{\lambda_1}{L - \mu} - \gamma^2 \end{bmatrix} \succcurlyeq 0$$
$$0 = \lambda_1 - \lambda_2.$$

◇ Weak duality: any dual feasible point $\equiv$ valid worst-case convergence rate

| Dual problem — back to gradient descent

◇ Dual problem is

$$\min_{\tau, \lambda_1, \lambda_2 \geq 0} \tau \quad \text{subject to} \quad S = \begin{bmatrix} \tau - 1 + \frac{\lambda_1 L \mu}{L - \mu} & \gamma - \frac{\lambda_1 (L + \mu)}{2(L - \mu)} \\ \gamma - \frac{\lambda_1 (L + \mu)}{2(L - \mu)} & \frac{\lambda_1}{L - \mu} - \gamma^2 \end{bmatrix} \succcurlyeq 0$$
$$0 = \lambda_1 - \lambda_2.$$

◇ Weak duality: any dual feasible point $\equiv$ valid worst-case convergence rate ($\Uparrow$):

$$\|x_1 - x_\star\|^2 \leq \tau \|x_0 - x_\star\|^2 \text{ for all } f \in \mathcal{F}_{\mu, L}, \text{ all } x_0 \in \mathbb{R}^d, \text{ all } d \in \mathbb{N}, \text{ with}$$
$$x_1 = x_0 - \gamma \nabla f(x_0).$$



$$\exists \lambda, \tau \geq 0 : \begin{bmatrix} \tau - 1 + \frac{\lambda L \mu}{L - \mu} & \gamma - \frac{\lambda (L + \mu)}{2(L - \mu)} \\ \gamma - \frac{\lambda (L + \mu)}{2(L - \mu)} & \frac{\lambda}{L - \mu} - \gamma^2 \end{bmatrix} \succcurlyeq 0$$

| Dual problem — back to gradient descent

- ◇ Dual problem is

$$\min_{\tau, \lambda_1, \lambda_2 \geq 0} \tau \quad \text{subject to} \quad S = \begin{bmatrix} \tau - 1 + \frac{\lambda_1 L \mu}{L - \mu} & \gamma - \frac{\lambda_1 (L + \mu)}{2(L - \mu)} \\ \gamma - \frac{\lambda_1 (L + \mu)}{2(L - \mu)} & \frac{\lambda_1}{L - \mu} - \gamma^2 \end{bmatrix} \succcurlyeq 0$$
$$0 = \lambda_1 - \lambda_2.$$

- ◇ Weak duality: any dual feasible point $\equiv$ valid worst-case convergence rate ($\Uparrow$):

$$\|x_1 - x_\star\|^2 \leq \tau \|x_0 - x_\star\|^2 \text{ for all } f \in \mathcal{F}_{\mu, L}, \text{ all } x_0 \in \mathbb{R}^d, \text{ all } d \in \mathbb{N}, \text{ with}$$
$$x_1 = x_0 - \gamma \nabla f(x_0).$$



$$\exists \lambda, \tau \geq 0 : \begin{bmatrix} \tau - 1 + \frac{\lambda L \mu}{L - \mu} & \gamma - \frac{\lambda (L + \mu)}{2(L - \mu)} \\ \gamma - \frac{\lambda (L + \mu)}{2(L - \mu)} & \frac{\lambda}{L - \mu} - \gamma^2 \end{bmatrix} \succcurlyeq 0$$

- ◇ Strong duality holds ($\exists$ Slater point): any valid worst-case convergence rate $\equiv$ valid dual feasible point ($\Downarrow$).

| Dual problem — back to gradient descent

- ◇ Dual problem is

$$\min_{\tau, \lambda_1, \lambda_2 \geq 0} \tau \quad \text{subject to} \quad S = \begin{bmatrix} \tau - 1 + \frac{\lambda_1 L \mu}{L - \mu} & \gamma - \frac{\lambda_1 (L + \mu)}{2(L - \mu)} \\ \gamma - \frac{\lambda_1 (L + \mu)}{2(L - \mu)} & \frac{\lambda_1}{L - \mu} - \gamma^2 \end{bmatrix} \succcurlyeq 0$$
$$0 = \lambda_1 - \lambda_2.$$

- ◇ Weak duality: any dual feasible point $\equiv$ valid worst-case convergence rate ($\Uparrow$):

$$\|x_1 - x_\star\|^2 \leq \tau \|x_0 - x_\star\|^2 \text{ for all } f \in \mathcal{F}_{\mu, L}, \text{ all } x_0 \in \mathbb{R}^d, \text{ all } d \in \mathbb{N}, \text{ with}$$
$$x_1 = x_0 - \gamma \nabla f(x_0).$$



$$\exists \lambda, \tau \geq 0 : \begin{bmatrix} \tau - 1 + \frac{\lambda L \mu}{L - \mu} & \gamma - \frac{\lambda (L + \mu)}{2(L - \mu)} \\ \gamma - \frac{\lambda (L + \mu)}{2(L - \mu)} & \frac{\lambda}{L - \mu} - \gamma^2 \end{bmatrix} \succcurlyeq 0$$

- ◇ Strong duality holds ($\exists$ Slater point): any valid worst-case convergence rate $\equiv$ valid dual feasible point ($\Downarrow$).

| Dual certificate for gradient descent

Guessed expression for the multipliers: does it yield valid proof?



Your turn! Complete and run the corresponding notebook cells.

| Dual problem, sums-of-squares, and standard proofs

◇ Primal problem (abstract form)

$$\begin{aligned} \max_{G, F} \langle b_{\text{obj}}, F \rangle + \text{Tr}(A_{\text{obj}} G) \quad \text{subject to:} \quad & \langle b_1, F \rangle + \text{Tr}(A_1 G) \leq 0 \\ & \langle b_2, F \rangle + \text{Tr}(A_2 G) \leq 0 \\ & \langle b_{\text{init}}, F \rangle + \text{Tr}(A_{\text{init}} G) = 1 \\ & G \succcurlyeq 0 \end{aligned}$$

| Dual problem, sums-of-squares, and standard proofs

◇ Primal problem (abstract form)

$$\begin{array}{ll} \max_{G, F} \langle b_{\text{obj}}, F \rangle + \text{Tr}(A_{\text{obj}} G) & \text{subject to:} \\ \langle b_1, F \rangle + \text{Tr}(A_1 G) \leq 0 & : \lambda_1 \geq 0 \\ \langle b_2, F \rangle + \text{Tr}(A_2 G) \leq 0 & : \lambda_2 \geq 0 \\ \langle b_{\text{init}}, F \rangle + \text{Tr}(A_{\text{init}} G) = 1 & : \tau \\ G \succcurlyeq 0 & : S \succcurlyeq 0 \end{array}$$

| Dual problem, sums-of-squares, and standard proofs

◇ Primal problem (abstract form)

$$\begin{aligned} \max_{G, F} \langle b_{\text{obj}}, F \rangle + \text{Tr}(A_{\text{obj}} G) \quad \text{subject to:} \quad & \langle b_1, F \rangle + \text{Tr}(A_1 G) \leq 0 & : \lambda_1 \geq 0 \\ & \langle b_2, F \rangle + \text{Tr}(A_2 G) \leq 0 & : \lambda_2 \geq 0 \\ & \langle b_{\text{init}}, F \rangle + \text{Tr}(A_{\text{init}} G) = 1 & : \tau \\ & G \succcurlyeq 0 & : S \succcurlyeq 0 \end{aligned}$$

◇ Dual problem (minimize upper bound)

$$\begin{aligned} \min_{\substack{\lambda_1, \lambda_2 \geq 0 \\ S \succcurlyeq 0 \\ \tau \in \mathbb{R}}} \tau \quad \text{subject to:} \quad & 0 = b_{\text{obj}} - \lambda_1 b_1 - \lambda_2 b_2 - \tau b_{\text{init}} \\ & S = \lambda_1 A_1 + \lambda_2 A_2 + \tau A_{\text{init}} - A_{\text{obj}} \end{aligned}$$

| Dual problem, sums-of-squares, and standard proofs

◇ Primal problem (abstract form)

$$\begin{aligned} \max_{G, F} \langle b_{\text{obj}}, F \rangle + \text{Tr}(A_{\text{obj}} G) \quad \text{subject to:} \quad & \langle b_1, F \rangle + \text{Tr}(A_1 G) \leq 0 & : \lambda_1 \geq 0 \\ & \langle b_2, F \rangle + \text{Tr}(A_2 G) \leq 0 & : \lambda_2 \geq 0 \\ & \langle b_{\text{init}}, F \rangle + \text{Tr}(A_{\text{init}} G) = 1 & : \tau \\ & G \succcurlyeq 0 & : S \succcurlyeq 0 \end{aligned}$$

◇ Dual problem (minimize upper bound)

$$\begin{aligned} \min_{\substack{\lambda_1, \lambda_2 \geq 0 \\ S \succcurlyeq 0 \\ \tau \in \mathbb{R}}} \tau \quad \text{subject to:} \quad & 0 = b_{\text{obj}} - \lambda_1 b_1 - \lambda_2 b_2 - \tau b_{\text{init}} \\ & S = \lambda_1 A_1 + \lambda_2 A_2 + \tau A_{\text{init}} - A_{\text{obj}} \end{aligned}$$

◇ Remark that: if $(\lambda_1, \lambda_2, \tau, S)$ dual feasible, then by construction, for all primal feasible (G, F) :

| Dual problem, sums-of-squares, and standard proofs

◇ Primal problem (abstract form)

$$\begin{aligned} \max_{G, F} \langle b_{\text{obj}}, F \rangle + \text{Tr}(A_{\text{obj}} G) \quad \text{subject to:} \quad & \langle b_1, F \rangle + \text{Tr}(A_1 G) \leq 0 & : \lambda_1 \geq 0 \\ & \langle b_2, F \rangle + \text{Tr}(A_2 G) \leq 0 & : \lambda_2 \geq 0 \\ & \langle b_{\text{init}}, F \rangle + \text{Tr}(A_{\text{init}} G) = 1 & : \tau \\ & G \succcurlyeq 0 & : S \succcurlyeq 0 \end{aligned}$$

◇ Dual problem (minimize upper bound)

$$\begin{aligned} \min_{\substack{\lambda_1, \lambda_2 \geq 0 \\ S \succcurlyeq 0 \\ \tau \in \mathbb{R}}} \tau \quad \text{subject to:} \quad & 0 = b_{\text{obj}} - \lambda_1 b_1 - \lambda_2 b_2 - \tau b_{\text{init}} \\ & S = \lambda_1 A_1 + \lambda_2 A_2 + \tau A_{\text{init}} - A_{\text{obj}} \end{aligned}$$

◇ Remark that: if $(\lambda_1, \lambda_2, \tau, S)$ dual feasible, then by construction, for all primal feasible (G, F) :

$$0 \geq \lambda_1 (\langle b_1, F \rangle + \text{Tr}(A_1 G)) + \lambda_2 (\langle b_2, F \rangle + \text{Tr}(A_2 G))$$

| Dual problem, sums-of-squares, and standard proofs

◇ Primal problem (abstract form)

$$\begin{aligned} \max_{G, F} \langle b_{\text{obj}}, F \rangle + \text{Tr}(A_{\text{obj}} G) \quad \text{subject to:} \quad & \langle b_1, F \rangle + \text{Tr}(A_1 G) \leq 0 & : \lambda_1 \geq 0 \\ & \langle b_2, F \rangle + \text{Tr}(A_2 G) \leq 0 & : \lambda_2 \geq 0 \\ & \langle b_{\text{init}}, F \rangle + \text{Tr}(A_{\text{init}} G) = 1 & : \tau \\ & G \succcurlyeq 0 & : S \succcurlyeq 0 \end{aligned}$$

◇ Dual problem (minimize upper bound)

$$\begin{aligned} \min_{\substack{\lambda_1, \lambda_2 \geq 0 \\ S \succcurlyeq 0 \\ \tau \in \mathbb{R}}} \tau \quad \text{subject to:} \quad & 0 = b_{\text{obj}} - \lambda_1 b_1 - \lambda_2 b_2 - \tau b_{\text{init}} \\ & S = \lambda_1 A_1 + \lambda_2 A_2 + \tau A_{\text{init}} - A_{\text{obj}} \end{aligned}$$

◇ Remark that: if $(\lambda_1, \lambda_2, \tau, S)$ dual feasible, then by construction, for all primal feasible (G, F) :

$$\begin{aligned} 0 &\geq \lambda_1 (\langle b_1, F \rangle + \text{Tr}(A_1 G)) + \lambda_2 (\langle b_2, F \rangle + \text{Tr}(A_2 G)) \\ &= \langle b_{\text{obj}}, F \rangle + \text{Tr}(A_{\text{obj}} G) - \tau (\langle b_{\text{init}}, F \rangle + \text{Tr}(A_{\text{init}} G)) + \text{Tr}(S G) \end{aligned}$$

| Dual problem, sums-of-squares, and standard proofs

◇ Primal problem (abstract form)

$$\begin{aligned} \max_{G, F} \quad & \langle b_{\text{obj}}, F \rangle + \text{Tr}(A_{\text{obj}} G) \quad \text{subject to:} \quad \langle b_1, F \rangle + \text{Tr}(A_1 G) \leq 0 & : \lambda_1 \geq 0 \\ & \langle b_2, F \rangle + \text{Tr}(A_2 G) \leq 0 & : \lambda_2 \geq 0 \\ & \langle b_{\text{init}}, F \rangle + \text{Tr}(A_{\text{init}} G) = 1 & : \tau \\ & G \succcurlyeq 0 & : S \succcurlyeq 0 \end{aligned}$$

◇ Dual problem (minimize upper bound)

$$\begin{aligned} \min_{\substack{\lambda_1, \lambda_2 \geq 0 \\ S \succcurlyeq 0 \\ \tau \in \mathbb{R}}} \quad & \tau \quad \text{subject to:} \quad 0 = b_{\text{obj}} - \lambda_1 b_1 - \lambda_2 b_2 - \tau b_{\text{init}} \\ & S = \lambda_1 A_1 + \lambda_2 A_2 + \tau A_{\text{init}} - A_{\text{obj}} \end{aligned}$$

◇ Remark that: if $(\lambda_1, \lambda_2, \tau, S)$ dual feasible, then by construction, for all primal feasible (G, F) :

$$\begin{aligned} 0 & \geq \lambda_1 (\langle b_1, F \rangle + \text{Tr}(A_1 G)) + \lambda_2 (\langle b_2, F \rangle + \text{Tr}(A_2 G)) \\ & = \underbrace{\langle b_{\text{obj}}, F \rangle + \text{Tr}(A_{\text{obj}} G)}_{\text{performance measure}} - \tau (\langle b_{\text{init}}, F \rangle + \text{Tr}(A_{\text{init}} G)) + \text{Tr}(S G) \end{aligned}$$

| Dual problem, sums-of-squares, and standard proofs

◇ Primal problem (abstract form)

$$\begin{aligned} \max_{G, F} \quad & \langle b_{\text{obj}}, F \rangle + \text{Tr}(A_{\text{obj}} G) \quad \text{subject to:} \quad \langle b_1, F \rangle + \text{Tr}(A_1 G) \leq 0 & : \lambda_1 \geq 0 \\ & \langle b_2, F \rangle + \text{Tr}(A_2 G) \leq 0 & : \lambda_2 \geq 0 \\ & \langle b_{\text{init}}, F \rangle + \text{Tr}(A_{\text{init}} G) = 1 & : \tau \\ & G \succcurlyeq 0 & : S \succcurlyeq 0 \end{aligned}$$

◇ Dual problem (minimize upper bound)

$$\begin{aligned} \min_{\substack{\lambda_1, \lambda_2 \geq 0 \\ S \succcurlyeq 0 \\ \tau \in \mathbb{R}}} \quad & \tau \quad \text{subject to:} \quad 0 = b_{\text{obj}} - \lambda_1 b_1 - \lambda_2 b_2 - \tau b_{\text{init}} \\ & S = \lambda_1 A_1 + \lambda_2 A_2 + \tau A_{\text{init}} - A_{\text{obj}} \end{aligned}$$

◇ Remark that: if $(\lambda_1, \lambda_2, \tau, S)$ dual feasible, then by construction, for all primal feasible (G, F) :

$$\begin{aligned} 0 & \geq \lambda_1 (\langle b_1, F \rangle + \text{Tr}(A_1 G)) + \lambda_2 (\langle b_2, F \rangle + \text{Tr}(A_2 G)) \\ & = \underbrace{\langle b_{\text{obj}}, F \rangle + \text{Tr}(A_{\text{obj}} G)}_{\text{performance measure}} - \underbrace{\tau (\langle b_{\text{init}}, F \rangle + \text{Tr}(A_{\text{init}} G))}_{\text{initial condition}} + \text{Tr}(S G) \end{aligned}$$

| Dual problem, sums-of-squares, and standard proofs

◇ Primal problem (abstract form)

$$\begin{aligned} \max_{G, F} \quad & \langle b_{\text{obj}}, F \rangle + \text{Tr}(A_{\text{obj}} G) \quad \text{subject to:} \quad \langle b_1, F \rangle + \text{Tr}(A_1 G) \leq 0 & : \lambda_1 \geq 0 \\ & \langle b_2, F \rangle + \text{Tr}(A_2 G) \leq 0 & : \lambda_2 \geq 0 \\ & \langle b_{\text{init}}, F \rangle + \text{Tr}(A_{\text{init}} G) = 1 & : \tau \\ & G \succeq 0 & : S \succeq 0 \end{aligned}$$

◇ Dual problem (minimize upper bound)

$$\begin{aligned} \min_{\substack{\lambda_1, \lambda_2 \geq 0 \\ S \succeq 0 \\ \tau \in \mathbb{R}}} \quad & \tau \quad \text{subject to:} \quad 0 = b_{\text{obj}} - \lambda_1 b_1 - \lambda_2 b_2 - \tau b_{\text{init}} \\ & S = \lambda_1 A_1 + \lambda_2 A_2 + \tau A_{\text{init}} - A_{\text{obj}} \end{aligned}$$

◇ Remark that: if $(\lambda_1, \lambda_2, \tau, S)$ dual feasible, then by construction, for all primal feasible (G, F) :

$$\begin{aligned} 0 & \geq \lambda_1 (\langle b_1, F \rangle + \text{Tr}(A_1 G)) + \lambda_2 (\langle b_2, F \rangle + \text{Tr}(A_2 G)) \\ & = \underbrace{\langle b_{\text{obj}}, F \rangle + \text{Tr}(A_{\text{obj}} G)}_{\text{performance measure}} - \underbrace{\tau (\langle b_{\text{init}}, F \rangle + \text{Tr}(A_{\text{init}} G))}_{\text{initial condition}} + \underbrace{\text{Tr}(S G)}_{\text{SOS}} \end{aligned}$$

| Dual problem, sums-of-squares, and standard proofs

Gradient with $\gamma = \frac{1}{L}$. Perform weighted sum of two inequalities

| Dual problem, sums-of-squares, and standard proofs

Gradient with $\gamma = \frac{1}{L}$. Perform weighted sum of two inequalities

$$f_0 \geq f_\star + \frac{1}{2L} \|\nabla f(x_0)\|^2 + \frac{\mu}{2(1-\mu/L)} \left\| x_0 - x_\star - \frac{1}{L} \nabla f(x_0) \right\|^2 \quad : \lambda_1$$

$$f_\star \geq f_0 + \langle \nabla f(x_0), x_\star - x_0 \rangle + \frac{1}{2L} \|\nabla f(x_0)\|^2 + \frac{\mu}{2(1-\mu/L)} \left\| x_0 - x_\star - \frac{1}{L} \nabla f(x_0) \right\|^2 \quad : \lambda_2$$

| Dual problem, sums-of-squares, and standard proofs

Gradient with $\gamma = \frac{1}{L}$. Perform weighted sum of two inequalities

$$f_0 \geq f_\star + \frac{1}{2L} \|\nabla f(x_0)\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_0 - x_\star - \frac{1}{L} \nabla f(x_0)\|^2 \quad : \lambda_1$$

$$f_\star \geq f_0 + \langle \nabla f(x_0), x_\star - x_0 \rangle + \frac{1}{2L} \|\nabla f(x_0)\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_0 - x_\star - \frac{1}{L} \nabla f(x_0)\|^2 \quad : \lambda_2$$

with $\lambda_1, \lambda_2 \geq 0$. Weighted sum can be reformulated as

| Dual problem, sums-of-squares, and standard proofs

Gradient with $\gamma = \frac{1}{L}$. Perform weighted sum of two inequalities

$$f_0 \geq f_\star + \frac{1}{2L} \|\nabla f(x_0)\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_0 - x_\star - \frac{1}{L} \nabla f(x_0)\|^2 \quad : \lambda_1 = 2\gamma(1 - \gamma\mu)$$

$$f_\star \geq f_0 + \langle \nabla f(x_0), x_\star - x_0 \rangle + \frac{1}{2L} \|\nabla f(x_0)\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_0 - x_\star - \frac{1}{L} \nabla f(x_0)\|^2 \quad : \lambda_2 = 2\gamma(1 - \gamma\mu)$$

with $\lambda_1, \lambda_2 \geq 0$. Weighted sum can be reformulated as

| Dual problem, sums-of-squares, and standard proofs

Gradient with $\gamma = \frac{1}{L}$. Perform weighted sum of two inequalities

$$f_0 \geq f_\star + \frac{1}{2L} \|\nabla f(x_0)\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_0 - x_\star - \frac{1}{L} \nabla f(x_0)\|^2 \quad : \lambda_1 = 2\gamma(1 - \gamma\mu)$$

$$f_\star \geq f_0 + \langle \nabla f(x_0), x_\star - x_0 \rangle + \frac{1}{2L} \|\nabla f(x_0)\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_0 - x_\star - \frac{1}{L} \nabla f(x_0)\|^2 \quad : \lambda_2 = 2\gamma(1 - \gamma\mu)$$

with $\lambda_1, \lambda_2 \geq 0$. Weighted sum can be reformulated as

$$\|x_1 - x_\star\|^2 \leq (1 - \gamma\mu)^2 \|x_0 - x_\star\|^2 - \underbrace{\gamma \frac{2-\gamma(L+\mu)}{L-\mu} \|\mu(x_0 - x_\star) - \nabla f(x_0)\|^2}_{\text{non-negative}},$$

| Dual problem, sums-of-squares, and standard proofs

Gradient with $\gamma = \frac{1}{L}$. Perform weighted sum of two inequalities

$$f_0 \geq f_\star + \frac{1}{2L} \|\nabla f(x_0)\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_0 - x_\star - \frac{1}{L} \nabla f(x_0)\|^2 \quad : \lambda_1 = 2\gamma(1 - \gamma\mu)$$

$$f_\star \geq f_0 + \langle \nabla f(x_0), x_\star - x_0 \rangle + \frac{1}{2L} \|\nabla f(x_0)\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_0 - x_\star - \frac{1}{L} \nabla f(x_0)\|^2 \quad : \lambda_2 = 2\gamma(1 - \gamma\mu)$$

with $\lambda_1, \lambda_2 \geq 0$. Weighted sum can be reformulated as

$$\|x_1 - x_\star\|^2 \leq (1 - \gamma\mu)^2 \|x_0 - x_\star\|^2 - \underbrace{\gamma \frac{2-\gamma(L+\mu)}{L-\mu} \|\mu(x_0 - x_\star) - \nabla f(x_0)\|^2}_{\geq 0},$$

| Dual problem, sums-of-squares, and standard proofs

Gradient with $\gamma = \frac{1}{L}$. Perform weighted sum of two inequalities

$$f_0 \geq f_\star + \frac{1}{2L} \|\nabla f(x_0)\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_0 - x_\star - \frac{1}{L} \nabla f(x_0)\|^2 \quad : \lambda_1 = 2\gamma(1 - \gamma\mu)$$

$$f_\star \geq f_0 + \langle \nabla f(x_0), x_\star - x_0 \rangle + \frac{1}{2L} \|\nabla f(x_0)\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_0 - x_\star - \frac{1}{L} \nabla f(x_0)\|^2 \quad : \lambda_2 = 2\gamma(1 - \gamma\mu)$$

with $\lambda_1, \lambda_2 \geq 0$. Weighted sum can be reformulated as

$$\begin{aligned} \|x_1 - x_\star\|^2 &\leq (1 - \gamma\mu)^2 \|x_0 - x_\star\|^2 - \underbrace{\gamma \frac{2-\gamma(L+\mu)}{L-\mu} \|\mu(x_0 - x_\star) - \nabla f(x_0)\|^2}_{\geq 0}, \\ &\leq (1 - \gamma\mu)^2 \|x_0 - x_\star\|^2, \end{aligned}$$

| Dual problem, sums-of-squares, and standard proofs

Gradient with $\gamma = \frac{1}{L}$. Perform weighted sum of two inequalities

$$f_0 \geq f_\star + \frac{1}{2L} \|\nabla f(x_0)\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_0 - x_\star - \frac{1}{L} \nabla f(x_0)\|^2 \quad : \lambda_1 = 2\gamma(1 - \gamma\mu)$$

$$f_\star \geq f_0 + \langle \nabla f(x_0), x_\star - x_0 \rangle + \frac{1}{2L} \|\nabla f(x_0)\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_0 - x_\star - \frac{1}{L} \nabla f(x_0)\|^2 \quad : \lambda_2 = 2\gamma(1 - \gamma\mu)$$

with $\lambda_1, \lambda_2 \geq 0$. Weighted sum can be reformulated as

$$\begin{aligned} \|x_1 - x_\star\|^2 &\leq (1 - \gamma\mu)^2 \|x_0 - x_\star\|^2 - \underbrace{\gamma \frac{2-\gamma(L+\mu)}{L-\mu} \|\mu(x_0 - x_\star) - \nabla f(x_0)\|^2}_{\geq 0}, \\ &\leq (1 - \gamma\mu)^2 \|x_0 - x_\star\|^2, \end{aligned}$$

leading to $\|x_1 - x_\star\|^2 \leq (1 - \gamma\mu)^2 \|x_0 - x_\star\|^2$

| Dual problem, sums-of-squares, and standard proofs

Gradient with $\gamma = \frac{1}{L}$. Perform weighted sum of two inequalities

$$f_0 \geq f_\star + \frac{1}{2L} \|\nabla f(x_0)\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_0 - x_\star - \frac{1}{L} \nabla f(x_0)\|^2 \quad : \lambda_1 = 2\gamma(1 - \gamma\mu)$$

$$f_\star \geq f_0 + \langle \nabla f(x_0), x_\star - x_0 \rangle + \frac{1}{2L} \|\nabla f(x_0)\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_0 - x_\star - \frac{1}{L} \nabla f(x_0)\|^2 \quad : \lambda_2 = 2\gamma(1 - \gamma\mu)$$

with $\lambda_1, \lambda_2 \geq 0$. Weighted sum can be reformulated as

$$\begin{aligned} \|x_1 - x_\star\|^2 &\leq (1 - \gamma\mu)^2 \|x_0 - x_\star\|^2 - \underbrace{\gamma \frac{2-\gamma(L+\mu)}{L-\mu} \|\mu(x_0 - x_\star) - \nabla f(x_0)\|^2}_{\geq 0, \text{ or } = 0 \text{ when worst-case is achieved}}, \\ &\leq (1 - \gamma\mu)^2 \|x_0 - x_\star\|^2, \end{aligned}$$

leading to $\|x_1 - x_\star\|^2 \leq (1 - \gamma\mu)^2 \|x_0 - x_\star\|^2$

| Dual problem, sums-of-squares, and standard proofs

Gradient with $\gamma = \frac{1}{L}$. Perform weighted sum of two inequalities

$$f_0 \geq f_\star + \frac{1}{2L} \|\nabla f(x_0)\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_0 - x_\star - \frac{1}{L} \nabla f(x_0)\|^2 \quad : \lambda_1 = 2\gamma(1 - \gamma\mu)$$

$$f_\star \geq f_0 + \langle \nabla f(x_0), x_\star - x_0 \rangle + \frac{1}{2L} \|\nabla f(x_0)\|^2 + \frac{\mu}{2(1-\mu/L)} \|x_0 - x_\star - \frac{1}{L} \nabla f(x_0)\|^2 \quad : \lambda_2 = 2\gamma(1 - \gamma\mu)$$

with $\lambda_1, \lambda_2 \geq 0$. Weighted sum can be reformulated as

$$\begin{aligned} \|x_1 - x_\star\|^2 &\leq (1 - \gamma\mu)^2 \|x_0 - x_\star\|^2 - \underbrace{\gamma \frac{2-\gamma(L+\mu)}{L-\mu} \|\mu(x_0 - x_\star) - \nabla f(x_0)\|^2}_{\geq 0, \text{ or } = 0 \text{ when worst-case is achieved}}, \\ &\leq (1 - \gamma\mu)^2 \|x_0 - x_\star\|^2, \end{aligned}$$

leading to $\|x_1 - x_\star\|^2 \leq (1 - \gamma\mu)^2 \|x_0 - x_\star\|^2$ (tight).

PEPs on an example: gradient descent

PEPs and proof structures

PEPs to design a (surprising?) algorithm

| Insight into proof structures from PEPs

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ (continuously differentiable) with $f \in \mathcal{F}$. Find $x_\star \in \mathbb{R}^d$ such that

$$f(x_\star) = \min_{x \in \mathbb{R}^d} f(x).$$

For example, assume $(x_\star = 0, f_\star = 0$ for readability):

| Insight into proof structures from PEPs

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ (continuously differentiable) with $f \in \mathcal{F}$. Find $x_\star \in \mathbb{R}^d$ such that

$$f(x_\star) = \min_{x \in \mathbb{R}^d} f(x).$$

For example, assume ($x_\star = 0$, $f_\star = 0$ for readability):

◇ Gradient-based method, e.g.: $x_k = x_{k-1} - \sum_{i=0}^{k-1} \gamma_{k,i} \nabla f(x_i)$,

| Insight into proof structures from PEPs

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ (continuously differentiable) with $f \in \mathcal{F}$. Find $x_\star \in \mathbb{R}^d$ such that

$$f(x_\star) = \min_{x \in \mathbb{R}^d} f(x).$$

For example, assume $(x_\star = 0, f_\star = 0$ for readability):

- ◇ Gradient-based method, e.g.: $x_k = x_{k-1} - \sum_{i=0}^{k-1} \gamma_{k,i} \nabla f(x_i)$,
- ◇ $\mathcal{F}$ can be described through nice linear/quadratic (interpolation) inequalities

| Insight into proof structures from PEPs

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ (continuously differentiable) with $f \in \mathcal{F}$. Find $x_\star \in \mathbb{R}^d$ such that

$$f(x_\star) = \min_{x \in \mathbb{R}^d} f(x).$$

For example, assume ($x_\star = 0$, $f_\star = 0$ for readability):

- ◇ Gradient-based method, e.g.: $x_k = x_{k-1} - \sum_{i=0}^{k-1} \gamma_{k,i} \nabla f(x_i)$,
- ◇ $\mathcal{F}$ can be described through nice linear/quadratic (interpolation) inequalities in terms of $F = [f_0, \dots, f_n]$ and $P = [x_0, \nabla f(x_0), \dots, \nabla f(x_n)]$ of the form $\text{Tr}(P^T P A_i) + b_i^T F \leq 0$.

| Insight into proof structures from PEPs

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ (continuously differentiable) with $f \in \mathcal{F}$. Find $x_\star \in \mathbb{R}^d$ such that

$$f(x_\star) = \min_{x \in \mathbb{R}^d} f(x).$$

For example, assume ($x_\star = 0$, $f_\star = 0$ for readability):

- ◇ Gradient-based method, e.g.: $x_k = x_{k-1} - \sum_{i=0}^{k-1} \gamma_{k,i} \nabla f(x_i)$,
- ◇ $\mathcal{F}$ can be described through nice linear/quadratic (interpolation) inequalities in terms of $F = [f_0, \dots, f_n]$ and $P = [x_0, \nabla f(x_0), \dots, \nabla f(x_n)]$ of the form $\text{Tr}(P^T P A_i) + b_i^T F \leq 0$.
- ◇ Performance criterion/initial condition linear in $(G = P^T P, F)$.

| Insight into proof structures from PEPs

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ (continuously differentiable) with $f \in \mathcal{F}$. Find $x_\star \in \mathbb{R}^d$ such that

$$f(x_\star) = \min_{x \in \mathbb{R}^d} f(x).$$

For example, assume $(x_\star = 0, f_\star = 0$ for readability):

- ◇ Gradient-based method, e.g.: $x_k = x_{k-1} - \sum_{i=0}^{k-1} \gamma_{k,i} \nabla f(x_i)$,
- ◇ $\mathcal{F}$ can be described through nice linear/quadratic (interpolation) inequalities in terms of $F = [f_0, \dots, f_n]$ and $P = [x_0, \nabla f(x_0), \dots, \nabla f(x_n)]$ of the form $\text{Tr}(P^T P A_i) + b_i^T F \leq 0$.
- ◇ Performance criterion/initial condition linear in $(G = P^T P, F)$.

Then, proofs of the fact “Perf $\leq \tau$ Init”

| Insight into proof structures from PEPs

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ (continuously differentiable) with $f \in \mathcal{F}$. Find $x_\star \in \mathbb{R}^d$ such that

$$f(x_\star) = \min_{x \in \mathbb{R}^d} f(x).$$

For example, assume $(x_\star = 0, f_\star = 0$ for readability):

- ◇ Gradient-based method, e.g.: $x_k = x_{k-1} - \sum_{i=0}^{k-1} \gamma_{k,i} \nabla f(x_i)$,
- ◇ $\mathcal{F}$ can be described through nice linear/quadratic (interpolation) inequalities in terms of $F = [f_0, \dots, f_n]$ and $P = [x_0, \nabla f(x_0), \dots, \nabla f(x_n)]$ of the form $\text{Tr}(P^T P A_i) + b_i^T F \leq 0$.
- ◇ Performance criterion/initial condition linear in $(G = P^T P, F)$.

Then, proofs of the fact “Perf $\leq \tau$ Init” can be framed as: $\exists (\lambda_i)_i, \lambda_i \geq 0$, and $S \succcurlyeq 0$

$$0 \geq \sum_i \lambda_i (\text{Tr}(G A_i) + b_i^T F) = \text{Perf} - \tau \text{Init} + \text{Tr}(G S).$$

| Gradient descent... again!

Toy example: What is the smallest τ such that:

$$f(x_1) - f_\star \leq \tau(f(x_0) - f_\star),$$

for all

| Gradient descent... again!

Toy example: What is the smallest τ such that:

$$f(x_1) - f_\star \leq \tau(f(x_0) - f_\star),$$

for all

- ◇ $d \in \mathbb{N}$, L -smooth and μ -strongly convex function f (notation $f \in \mathcal{F}_{\mu,L}$),
- ◇ x_0 , and x_1 generated by gradient step $x_1 = x_0 - \gamma \nabla f(x_0)$,
- ◇ $x_\star = \operatorname{argmin}_x f(x)$?

| Gradient descent... again!

Toy example: What is the smallest τ such that:

$$f(x_1) - f_\star \leq \tau(f(x_0) - f_\star),$$

for all

- ◇ $d \in \mathbb{N}$, L -smooth and μ -strongly convex function f (notation $f \in \mathcal{F}_{\mu,L}$),
- ◇ x_0 , and x_1 generated by gradient step $x_1 = x_0 - \gamma \nabla f(x_0)$,
- ◇ $x_\star = \underset{x}{\operatorname{argmin}} f(x)$?

Wooclap time!

<https://app.wooclap.com/THIKNZU?from=instruction-slide>



PEPs on an example: gradient descent

PEPs and proof structures

PEPs to design a (surprising?) algorithm

| Designing algorithms

A “generic” first-order method

$$w_1 = w_0 - \gamma_{1,0} \nabla f(w_0)$$

$$w_2 = w_1 - \gamma_{2,0} \nabla f(w_0) - \gamma_{2,1} \nabla f(w_1)$$

$$w_3 = w_2 - \gamma_{3,0} \nabla f(w_0) - \gamma_{3,1} \nabla f(w_1) - \gamma_{3,2} \nabla f(w_2)$$

$\vdots$

$$w_n = w_{n-1} - \sum_{i=0}^{n-1} \gamma_{n,i} \nabla f(w_i),$$

(FOM)

for some coefficients $\{\gamma_{i,j}\}$. Generic **non-adaptive** first-order method.

| Designing algorithms

A “generic” first-order method

$$w_1 = w_0 - \gamma_{1,0} \nabla f(w_0)$$

$$w_2 = w_1 - \gamma_{2,0} \nabla f(w_0) - \gamma_{2,1} \nabla f(w_1)$$

$$w_3 = w_2 - \gamma_{3,0} \nabla f(w_0) - \gamma_{3,1} \nabla f(w_1) - \gamma_{3,2} \nabla f(w_2)$$

$\vdots$

$$w_n = w_{n-1} - \sum_{i=0}^{n-1} \gamma_{n,i} \nabla f(w_i),$$

(FOM)

for some coefficients $\{\gamma_{i,j}\}$. Generic **non-adaptive** first-order method.

How to choose $\{\gamma_{i,j}\}$?

| Designing algorithms

A “generic” first-order method

$$w_1 = w_0 - \gamma_{1,0} \nabla f(w_0)$$

$$w_2 = w_1 - \gamma_{2,0} \nabla f(w_0) - \gamma_{2,1} \nabla f(w_1)$$

$$w_3 = w_2 - \gamma_{3,0} \nabla f(w_0) - \gamma_{3,1} \nabla f(w_1) - \gamma_{3,2} \nabla f(w_2)$$

$\vdots$

$$w_n = w_{n-1} - \sum_{i=0}^{n-1} \gamma_{n,i} \nabla f(w_i),$$

(FOM)

for some coefficients $\{\gamma_{i,j}\}$. Generic **non-adaptive** first-order method.

How to choose $\{\gamma_{i,j}\}$?

- ◇ pick a performance criterion, for instance $\frac{\|w_n - w_\star\|^2}{\|w_0 - w_\star\|^2}$,

| Designing algorithms

A “generic” first-order method

$$\begin{aligned}w_1 &= w_0 - \gamma_{1,0} \nabla f(w_0) \\w_2 &= w_1 - \gamma_{2,0} \nabla f(w_0) - \gamma_{2,1} \nabla f(w_1) \\w_3 &= w_2 - \gamma_{3,0} \nabla f(w_0) - \gamma_{3,1} \nabla f(w_1) - \gamma_{3,2} \nabla f(w_2) \\&\vdots \\w_n &= w_{n-1} - \sum_{i=0}^{n-1} \gamma_{n,i} \nabla f(w_i),\end{aligned}\tag{FOM}$$

for some coefficients $\{\gamma_{i,j}\}$. Generic **non-adaptive** first-order method.

How to choose $\{\gamma_{i,j}\}$?

- ◇ pick a performance criterion, for instance $\frac{\|w_n - w_\star\|^2}{\|w_0 - w_\star\|^2}$,
- ◇ solve the minimax (minimize worst-case): $\min_{\{\gamma_{i,j}\}_{i,j}} \max_{f \in \mathcal{F}, \{w_i\}} \frac{\|w_n - w_\star\|^2}{\|w_0 - w_\star\|^2}.$

| Hands-on

Design an algorithm for minimizing
 L -smooth μ -strongly convex functions.

| Hands-on

Design an algorithm for minimizing
 L -smooth μ -strongly convex functions.



Your turn! Complete and run the corresponding notebook cells.

| Hands-on

Design an algorithm for minimizing
 L -smooth μ -strongly convex functions.



Your turn! Complete and run the
corresponding notebook cells.

◇ Goal 1: design an optimized one-step gradient method

$$x_1 = x_0 - \gamma_0 \nabla f(x_0),$$

for optimizing the worst-case performance $\frac{\|x_1 - x_\star\|^2}{\|x_0 - x_\star\|^2}$.

| Hands-on

Design an algorithm for minimizing
 L -smooth μ -strongly convex functions.



Your turn! Complete and run the
corresponding notebook cells.

- ◇ Goal 1: design an optimized one-step gradient method

$$x_1 = x_0 - \gamma_0 \nabla f(x_0),$$

for optimizing the worst-case performance $\frac{\|x_1 - x_\star\|^2}{\|x_0 - x_\star\|^2}$.

- ◇ Goal 2: design an optimized two-step gradient method²

$$x_1 = x_0 - \gamma_0 \nabla f(x_0),$$

$$x_2 = x_1 - \gamma_1 \nabla f(x_1),$$

for optimizing the worst-case performance $\frac{\|x_2 - x_\star\|^2}{\|x_0 - x_\star\|^2}$.

| Hands-on

Design an algorithm for minimizing
 L -smooth μ -strongly convex functions.



Your turn! Complete and run the
corresponding notebook cells.

- ◇ Goal 1: design an optimized one-step gradient method

$$x_1 = x_0 - \gamma_0 \nabla f(x_0),$$

for optimizing the worst-case performance $\frac{\|x_1 - x_\star\|^2}{\|x_0 - x_\star\|^2}$.

- ◇ Goal 2: design an optimized two-step gradient method²

$$x_1 = x_0 - \gamma_0 \nabla f(x_0),$$

$$x_2 = x_1 - \gamma_1 \nabla f(x_1),$$

for optimizing the worst-case performance $\frac{\|x_2 - x_\star\|^2}{\|x_0 - x_\star\|^2}$.

²Altschuler, Parrilo (2025). "Acceleration by Step-size Hedging I: Multi-Step Descent and the Silver Step-size Schedule"

| Concluding remarks

Performance estimation's philosophy

| Concluding remarks

Performance estimation's philosophy

- ◇ numerically allows obtaining **tight bounds** (rigorous baselines),
 - fast prototyping
 - worth checking before trying to prove a method works.

| Concluding remarks

Performance estimation's philosophy

- ◇ numerically allows obtaining **tight bounds** (rigorous baselines),
 - fast prototyping
 - worth checking before trying to prove a method works.
- ◇ algebraic insights into performance analyses:
 - analyses are dual feasible points,
 - analyses are linear combinations of certain specific inequalities.

| Concluding remarks

Performance estimation's philosophy

- ◇ numerically allows obtaining **tight bounds** (rigorous baselines),
 - fast prototyping
 - worth checking before trying to prove a method works.
- ◇ algebraic insights into performance analyses:
 - analyses are dual feasible points,
 - analyses are linear combinations of certain specific inequalities.

Byproducts:

- ◇ computer-assisted design of analyses,
- ◇ computer-assisted design of numerical methods,
- ◇ step towards reproducible theory: validation & benchmark tool for proofs!